

WEEK 1 INDEPENDENT EXERCISE

GitHub Workflow Practice

Estimated time: 20-25 minutes

Goal: Complete the full GitHub workflow independently: copy the template file, edit it, review your changes, commit, push, and verify changes on GitHub.

Before you begin

- ☐ Open the course-hub Rproject and pull the latest instructor updates.
- ☐ Locate the Week 1 exercise file in course-hub/week-01/exercises/.
- ☐ Confirm that your personal tutorials-exercises repository is beside course-hub, not inside it.

Part 1. Complete the workflow

1

Copy the template file. Copy the content of the exercise-01.R file (or copy the file itself) into tutorials-exercises/week-01/exercises/. Do not complete or save your work in course-hub.

2

Open the correct Rproject. Open tutorials-exercises.Rproj and confirm that tutorials-exercises appears as the active project in the upper-left corner of RStudio.

3

Complete the R script. Replace the placeholder text, run the code, and make sure the introduction object prints correctly.

```
# Week 1: GitHub workflow practice

# 1. In your own words, explain the difference between:
#   a. clone and pull
#   b. commit and push

# Write your answers below.

# 2. Create two R objects.
student_name <- "replace this with your name"
favorite_number <- "replace this with the number"

# 3. Use the objects to create a sentence.
introduction <- paste(
  "My name is",
  student_name,
  "and my favorite number is",
  favorite_number
)

introduction
```

4

Review your changes. Save the file. In the Git pane, stage the file, click Diff, and confirm that you changed only the intended file.

5

Commit the file. Use an informative commit message, such as: complete Week 1 GitHub exercise.

6

Push and verify. Push the commit, open your tutorials-exercises repository in a browser, refresh the page, and confirm that the file and latest commit appear online.

Part 2. Make a second commit

Open `exercise-01.R` and add the following reflection at the bottom of the file:

```
# Name one part of the GitHub workflow and how it fits into the bigger picture:
```

Write your answer below the comment. Then repeat the complete workflow:

Save

Review

Stage

Commit

Push

Verify

Use this commit message: `add GitHub workflow reflection`

Part 3. Troubleshooting challenge

Identify the most likely missing or incorrect step for each scenario.

- A. My file is saved in RStudio, but it does not appear on GitHub.
- B. I made a commit, but GitHub still shows the older version of the file.
- C. I opened RStudio and cannot find the Week 1 exercise.
- D. I pulled `course-hub`, but the instructor's new file is not visible.

Completion checklist for exercises-01

1. The exercise file is in `tutorials-exercises/week-01/exercises/`.
2. I edited and ran the R code in the `tutorials-exercises` project.
3. I reviewed the Diff before committing.
4. I created two informative commits.
5. I pushed both commits to GitHub.
6. I verified the file and commit history in the remote repository.

Remember: A saved file is not necessarily committed, and a committed file is not necessarily pushed.